SGA 4, Exposé V. Cohomology in Topoi

Draft translation, Introduction and Section 0

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Translator's status note. This fresh-only continuation starts Exposé V. The translated range corresponds to source lines 57–369 of the Orgogozo–Laszlo TeX snapshot. Terminology is normalized to current mathematical English: *topos*, *ringed topos*, *flat module*, *spectral sequence*, *derived category*, *sheaf cohomology*, *Čech cohomology*, *hypercovering*, *abelian category*, *injective object*, *generator*, *exact filtered inductive limits*, and *cofinal/cogenerating family* in the usual modern sense. Numbering follows the source.

Introduction

In this exposé we present the elementary commutative cohomological invariants of topoi. In no. 1, flat modules and flat morphisms of ringed topoi are studied. The proofs are given using the hypothesis, most often satisfied in practice, that the topoi have enough points (IV, 6). These proofs are taken up again in the general case in Appendix no. 8, where Deligne, using the technique of local inductive limits, also generalizes to the case of topoi D. Lazard's theorem on the structure of flat modules. The theorems of this exposé are existence theorems for spectral sequences relating the various cohomological invariants (nos. 3, 5, 6). It is known that, even for topological spaces, Čech cohomology does not in general coincide with sheaf cohomology [11]. In Appendix no. 7 a modified Čech calculation is introduced which makes it possible to obtain, by means of coverings, sheaf cohomology in an arbitrary topos. In that appendix one is led to use simplicial coverings, or hypercoverings, whose homotopy invariants were studied in [1] (cf. also [17]).

The cohomological invariants introduced here are elementary in the sense that derived categories are not used [12]. The reader familiar with that language will immediately translate the various statements of this exposé and will then be able to generalize them to complexes and to hypercohomology. The language of derived categories is, moreover, used later in this seminar.

We restrict ourselves here to commutative cohomology. For non-commutative H^1, used in this seminar, and for non-commutative H^2, we refer to [9]. The functors that are derived are additive; multiplicative structures are not discussed (cf. [7]).

0 Generalities on abelian categories

In this section we recall a few lemmas, most of which may be found in [11].

Proposition 0.1. Let $\mathfrak{A}$ be an abelian category having a generator. The following conditions are equivalent:

(i) The category $\mathfrak{A}$ satisfies axiom $AB5$): small direct sums are representable, and if $(X_i)_{i\in I}$ is a small increasing filtered family of subobjects of an object X of $\mathfrak{A}$, and if Y is a subobject of X, then

$$\left(\sup_i X_i\right) \cap Y = \sup_i (X_i \cap Y).$$

(ii) Small pseudo-filtered inductive limits *(I, 2.7)* are representable and commute with finite projective limits.

Moreover, if these conditions are satisfied, small filtered inductive limits are universal *(I, 2.6)*.

Proof. It is clear that (ii) implies (i). To show that (i) implies (ii), and to prove the supplementary assertion, it is enough to use the fact that $\mathfrak{A}$ is a full subcategory of a category $\mathcal{M}$ of modules over a suitable ring, in such a way that the inclusion functor

$$u : \mathfrak{A} \longrightarrow \mathcal{M}$$

has an exact left adjoint v [5]. The verification is then trivial.

0.1.1

It is known [11] that an abelian category $\mathfrak{A}$ having a generator and satisfying axiom $AB5$) has enough injectives, i.e. every object embeds into an injective object. Moreover, by the result already cited [5], small products are representable in $\mathfrak{A}$ (axiom $AB3$)*).

Proposition 0.2. Let $\mathfrak{A}$ and $\mathfrak{B}$ be two abelian categories, and let

$$\mathfrak{A} \underset{u}{\overset{v}{\rightleftarrows}} \mathfrak{B}$$

be two adjoint functors, with u left adjoint to v. Consider the two properties:

(i) the functor u is exact;

(ii) the functor v carries injective objects of $\mathfrak{B}$ to injective objects of $\mathfrak{A}$.

1) One always has the implication (i) $\Rightarrow$ (ii). If, moreover, every non-zero object of $\mathfrak{B}$ is the source of a non-zero morphism into an injective object, then (ii) $\Leftrightarrow$ (i).

2) Suppose that:

   a) the category $\mathfrak{B}$ has enough injectives;
   b) one of the two equivalent conditions (i) and (ii) above is satisfied;
   c) the functor u is faithful.

   Then the category $\mathfrak{A}$ has enough injectives.

Proof. The proof is left to the reader.

Remark 0.2.1. The category of abelian groups has enough injectives. Applying Proposition 0.2, one deduces that every category of unital modules over a unital ring has enough injectives. Applying then the result of [5] used in the proof of Proposition 0.1, together with Proposition 0.2, one deduces that every abelian category having a generator and exact small filtered inductive limits has enough injectives. This gives a new proof of that fact.

Proposition 0.3. Let $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$ be three abelian categories, and let

$$u : \mathfrak{A} \longrightarrow \mathfrak{B}, \qquad v : \mathfrak{B} \longrightarrow \mathfrak{C}$$

be two additive left exact functors. Suppose that $\mathfrak{A}$ and $\mathfrak{B}$ have enough injective objects. The following two properties are equivalent:

(i) There exists a spectral functor whose $E_2^{p,q}$-term is

$$R^p v\, R^q u$$

and which abuts to $R^{p+q}(vu)$, suitably filtered.

(ii) The functor u sends injective objects of $\mathfrak{A}$ to objects that are acyclic for the functor v.

Proof. The implication (i) ⇒ (ii) is trivial, since it is enough to apply the spectral functor to an injective object. The implication (ii) ⇒ (i) is proved in [11].

Proposition 0.4. Let $\mathfrak{A}$ and $\mathfrak{B}$ be two abelian categories, and let

$$u : \mathfrak{A} \longrightarrow \mathfrak{B}$$

be an additive left exact functor. Let M be a subset of the set of objects of $\mathfrak{A}$ having the following properties:

1) Every object of $\mathfrak{A}$ embeds into an element of M.

2) If $X \oplus Y$ belongs to M, then X belongs to M.

3) If

$$0 \longrightarrow X' \longrightarrow X \longrightarrow X'' \longrightarrow 0$$

is an exact sequence and if X' and X belong to M, then X'' belongs to M and the sequence

$$0 \longrightarrow u(X') \longrightarrow u(X) \longrightarrow u(X'') \longrightarrow 0$$

is exact. The zero objects belong to M.

Then every injective object belongs to M, and the objects of M are acyclic for u, i.e. for every $q \neq 0$ and every object X of M one has $R^q u(X) = 0$. In particular, resolutions by objects of M allow one to compute the derived functors of u.

For the proof, see [11], 3.3.1.

Proposition 0.5. Let $\mathcal{U} \subset \mathcal{V}$ be two universes. Let $\mathfrak{A}$ (respectively $\mathfrak{B}$) be a $\mathcal{U}$-category (respectively a $\mathcal{V}$-category) which is abelian, satisfies axiom $AB5$) relative to $\mathcal{U}$ (respectively to $\mathcal{V}$), and has a $\mathcal{U}$-small (respectively $\mathcal{V}$-small) generating family. Let

$$\epsilon : \mathfrak{A} \longrightarrow \mathfrak{B}$$

be an exact fully faithful functor. The following conditions are equivalent:

1) There exists a generating family $(X_i)_{i\in I}$ of $\mathfrak{A}$ such that the family $(\epsilon(X_i))_{i\in I}$ is generating in $\mathfrak{B}$.

1′) Every object of $\mathfrak{B}$ is isomorphic to a quotient of an object of the form

$$\bigoplus_{\alpha\in A} \epsilon(Y_\alpha), \qquad A \in \mathcal{V}.$$

Under these conditions, one has the following property:

2) ϵ sends $\mathcal{U}$-small products to products, and hence commutes with $\mathcal{U}$-small projective limits.

Moreover, under the equivalent conditions 1) or 1′), the following conditions are equivalent:

a) For every object Y of $\mathfrak{A}$, every subobject of $\epsilon(Y)$ is isomorphic to the image under ϵ of a subobject of Y.

a′) There exists a generating family $(X_i)_{i\in I}$ of $\mathfrak{A}$ such that the family $(\epsilon(X_i))_{i\in I}$ is generating in $\mathfrak{B}$, and such that, for every $i \in I$, every subobject of $\epsilon(X_i)$ is isomorphic to the image under ϵ of a subobject of X_i.

b) Every object of $\mathfrak{B}$ is isomorphic to a subobject of an object of the form

$$\prod_{\alpha\in A} \epsilon(Y_\alpha), \qquad A \in \mathcal{V}.$$

c) ϵ commutes with $\mathcal{U}$-small direct sums, and hence commutes with $\mathcal{U}$-small inductive limits.

Finally, under conditions 1) and a′), one has:

d) ϵ sends injective objects to injective objects.

Remark 0.5.1. When in Proposition 0.5 one has $\mathcal{U} = \mathcal{V}$, conditions 1) and a′) imply that ϵ is an equivalence of categories, since one then has b), 2), and a).

0.5.2

We shall give only indications for the proof. The implications 1) $\Leftrightarrow$ 1′) and 1) $\Rightarrow$ 2) are left to the reader. It is clear that a) implies a′). Let us show that a′) implies d). Let M be an injective object of $\mathfrak{A}$. For every $i \in I$ and every subobject $Y \hookrightarrow X_i$ of X_i, the homomorphism

$$\mathrm{Hom}(X_i, M) \longrightarrow \mathrm{Hom}(Y, M)$$

is surjective. Hence, by a′) and by the full faithfulness of ϵ, for every subobject $U \hookrightarrow \epsilon(X_i)$ the homomorphism

$$\mathrm{Hom}(\epsilon(X_i), \epsilon(M)) \longrightarrow \mathrm{Hom}(U, \epsilon(M))$$

is surjective. Since the $\epsilon(X_i)$ form a generating family of $\mathfrak{B}$, the object $\epsilon(M)$ is injective [11].

Let us show that a′) implies b). Enlarging the family of the X_i if necessary, one may suppose that for every $i \in I$ every quotient of X_i is isomorphic to some X_j. Let, for every $i \in I$, $X_i \hookrightarrow M_i$ be a monomorphism into an injective object. The family $(\epsilon(X_i))_{i\in I}$ is stable under quotients, and for $i \in I$ the morphism

$$\epsilon(X_i) \hookrightarrow \epsilon(M_i)$$

is, by d), a monomorphism into an injective object. One then checks immediately that, since the family $\epsilon(X_i)$ is generating, the family $(\epsilon(M_i))_{i\in I}$ is cogenerating; hence b).

Let us show that b) implies c). Let $(Z_\alpha)_{\alpha\in A}$ be a $\mathcal{U}$-small family of objects of $\mathfrak{A}$, and let us prove that the canonical morphism

$$\bigoplus_{\alpha\in A} \epsilon(Z_\alpha) \longrightarrow \epsilon\left(\bigoplus_{\alpha\in A} Z_\alpha\right)$$

is an isomorphism. Since the family of objects $\epsilon(Y)$ is cogenerating, it is enough to show that for every object Y of $\mathfrak{A}$ the homomorphism

$$\operatorname{Hom}\left(\epsilon\left(\bigoplus_{\alpha} Z_\alpha\right), \epsilon(Y)\right) \longrightarrow \operatorname{Hom}\left(\bigoplus_{\alpha} \epsilon(Z_\alpha), \epsilon(Y)\right)$$

is an isomorphism, which follows from the full faithfulness of ϵ.

It remains to show that c) implies a). Let Z be a subobject of $\epsilon(Y)$. By $1'$), there exists a $\mathcal{U}$-small family Y_α of objects of $\mathfrak{A}$ and an epimorphism from $\bigoplus_\alpha \epsilon(Y_\alpha)$ onto Z. Since ϵ commutes with $\mathcal{U}$-small direct sums, there is therefore an epimorphism

$$\epsilon(Y') \longrightarrow Z.$$

Let $u : \epsilon(Y') \to Z \to \epsilon(Y)$ be the composite morphism. One has $Z = \operatorname{Im}(u)$. Since ϵ is fully faithful, one has $u = \epsilon(v)$, and consequently

$$Z = \operatorname{Im}(\epsilon(v)) = \epsilon(\operatorname{Im}(v)).$$

Exercise 0.5.2. Let $\mathcal{S}ex_{\mathcal{V}}(\mathfrak{A})$ be the category of contravariant functors from $\mathfrak{A}$ to the category of $\mathcal{V}$-abelian groups which commute with $\mathcal{U}$-small inductive limits. Show that under conditions 1) and a$'$) the canonical functor from $\mathfrak{B}$ to $\mathcal{S}ex_{\mathcal{V}}(\mathfrak{A})$ is an equivalence.